

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE CODE: CSA1402

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 20

Total Marks:100

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%				
Explanation	25%				
Application	20%				
Organization	15%				
Accuracy	10%				
Clarity	5%				

Faculty Incharge

SHORT ANSWER TEST

1. Construct language for the following Regular Expression:2

$$(a|b)^*a(a|b)|(a|b)$$

2. a) Illustrate the predictive parser for the following grammar. 4

$$S \rightarrow (L) | a$$

$$L \rightarrow L, S | S$$

3. Analyse, is it possible, by modifying the grammar in any way to construct predictive parser for the language of

$$S \rightarrow SS+ | SS^* | a$$

String: "aa+a" 4

4. Eliminate left recursion from the grammar $S \rightarrow Sa|Sb|c$ 2
 5. The following grammar contains common prefixes, making it unsuitable for predictive parsing.

$$A \rightarrow abcD | abcE | f$$

Apply left factoring to remove the common prefixes and rewrite the grammar.5

6. Consider the following statement:10

$$Degree - F = Degree - C * 1.8 + 32$$

Show the representation of this statement after each phase of the compiler and explain.

7. Consider the following C code fragment:4

```
sum = 0;
```

```
for (i = 1; i < 5; i++)
    sum = sum + i;
```

Show the representation of the above code after each phase of the compiler.2

8. Consider the following grammar.

$$S \rightarrow aABb, A \rightarrow c|\epsilon, B \rightarrow d|\epsilon$$

Calculate FIRST and FOLLOW, Obtain the Predictive Parsing Table.

not. Show the parser moves for the input string **acdb**.

9. Write a simple Lex program to recognize and count identifiers in a given input.2

10. Write the rule to Eliminate Left Recursion and Left Factoring.2

11. What are the actions performed by a shift-reduce parser? Explain each action with a suitable example.3

12. Consider the following grammar:

$$S \rightarrow pBB \rightarrow qB|r$$

Find the leftmost derivation and construct the parse tree for the input string pqqrr.3

13. Consider Grammar: $E \rightarrow E+T|T$ $T \rightarrow T*F|F$ $F \rightarrow (E)|id$

Eliminate left recursion and find FIRST and FOLLOW 5

14. **For the following grammar, determine whether the string abcc is accepted using shift-reduce parsing. Show all shift and reduce operations.**

$$S \rightarrow ABA \rightarrow aA | aB \rightarrow cB | c$$

15. For the following grammar, verify whether the string **abb**c is valid using shift-reduce parsing.10

$$S \rightarrow aABA \rightarrow bA | bB \rightarrow c$$

Show the complete sequence of stack, input, and parser actions.

16. Consider the following grammar:10

$$S' \rightarrow SS \rightarrow aABA \rightarrow bA | cB \rightarrow d$$

- Construct the canonical collection of LR(0) items.
- Construct the SLR(1) parsing table.

17. Consider the following grammar.

$$S \rightarrow iEtSS' | a, S' \rightarrow eS | \epsilon, E \rightarrow b$$

Obtain the Predictive Parsing Table. 5

18. Do left factoring in the following grammar: 3

$$A \rightarrow aAB | aA | a$$

$$B \rightarrow bB | b$$

19. Construct the LL(1) Parsing Table for the following grammars.7

$$S \rightarrow ABCA \rightarrow a | \epsilon B \rightarrow r | \epsilon C \rightarrow b | \epsilon$$

20. Construct the LL(1) Parsing Table for the following grammars.7

$$S \rightarrow 1S\# | qABCA \rightarrow a | bbDB \rightarrow a | \epsilon C \rightarrow b | \epsilon D \rightarrow c | \epsilon$$

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand